Lab 4 Solutions Group(9-16)

Secret Project :

```
lst = [int(x) for x in input().split()]
for i in lst:
    if i % 2 == 0:
        print('1',end='')
    else:
        print('0',end='')
```

Missing Student :

```
lst = [int(x) for x in input().split()]
arr = [i for i in range(len(lst)+1)]
for i in arr:
    if i not in lst:
        print(i)
        break
```

Email Validation :

```
n = int(input())
lst = []
for i in range(n):

    string = input()
    if string.count('.') != 1 or string.count("@")!= 1:
        continue
    if string.index(".") < string.index("@"):
        continue
    isAlpha = False;
    isDigit = False
    isAfter =False;
    isXtra = False
    for j in range(len(string)):
        if(string[j] >='0' and string[j] <='9'):
            isDigit = True;
            if j > string.index("@"):
```

```
                isAfter = True
        elif((string[j] >='a' and string[j] <='z') or (string[j] >='A' and string[j] <='Z')):
            isAlpha = True;
        elif string[j] =='@' or string[j] == '.':
            continue
        else :
            isXtra = True
    if isXtra or isAfter or not(isDigit) or not(isAlpha):
        continue
    lst.append(string)
print(len(lst))
for x in lst:
    print(x)
```

Polygon?

```
# Python 3 implementation of the approach

# Function to check whether
# it is possible to create a
# polygon with given sides length
def isPossible(a, n):
    # Sum stores the sum of all the sides
    # and maxS stores the length of
    # the largest side
    sum = 0
    maxS = 0
    for i in range(n):
        sum += a[i]
        maxS = max(a[i], maxS)

    # If the length of the largest side
    # is less than the sum of the
    # other remaining sides
    if ((sum - maxS) > maxS):
        return True

    return False


n = int(input())
a = []
```

```
for i in range(n):
    a.append(int(input()))

print(isPossible(a, n))
```

Count Vowels

```
s = input()

arr = [0, 0, 0, 0, 0]

for i in s:
    if i == 'a':
        arr[0] += 1
    elif i == 'e':
        arr[1] += 1
    elif i == 'i':
        arr[2] += 1
    elif i == 'o':
        arr[3] += 1
    elif i == 'u':
        arr[4] += 1


print('a:', arr[0])
print('e:', arr[1])
print('i:', arr[2])
print('o:', arr[3])
print('u:', arr[4])
```

Help Sam Again !!

```
string = input()
n = len(string)
mx = 0
for i in range(n):
    f = string[i]
    s = string[i]
```

```
    j = i
    while j<n :
        if(string[j]==f):
            j+=1;
            continue
        if(f==s):
            s = string[j];
            j+=1;
            continue;
        if(string[j]==s):
            j+=1;
            continue;
        break;
        j+=1;
    mx = max(mx,j-i)
print(mx)
```

Tom and Strings

```
l = input().split()
j = 0
ans = 0
while(True):
    flag = True
    if(len(l[0])<=j):
        break
    s = l[0][j]
    for i in range(1,len(l)):
        if(len(l[i])<=j or l[i][j]!=s):
            flag = False
            break
    if(flag):
        ans+=1
    else:
        break
    j+=1
print(ans)
```